Impact of Monetary Policy Change on Bank Risk

Dimitrije Ankic

29/08/2025

Contents

Introduction 3
Literature Review and Hypothesis Development 3
Methodology 4
Analysis of Results 5
Recommendations 10
Conclusion 10
References 12

Introduction

As a result of the changing global economy, as well as world-wide events such as the COVID19 pandemic, the relationship, and the means by which monetary policy changes affect financial institutions has garnered increasing attention. With the Australian banking sector being dominated by four banks: ANZ, NAB, Westpac, and CBA, aptly named "The Big Four", understanding how changes in monetary policy affect a bank's level of risk-aversion is essential for policy makers, financial institution workers, and the everyday Australian. The following report aims to discern the effects of the cash rate level on a bank's willingness to take risk by utilizing a data driven approach as a means of isolating the effects of monetary policy from other macroeconomic factors. Relevant literature was reviewed to better ascertain the correlation between the two factors, as well as to evaluate the results of previous studies. Based on this, a null hypothesis stating that the risks banks undertake are not correlated with monetary policy was formed. An Ordinary Least Squares (OLS) regression was combined with other visual analytics to evaluate the effects of monetary policy on bank risk, which was measured in a standardized Z-score form, the data used in the study was provided by RMIT University. The findings of the study contradicted the hypothesis, and further accentuate the link between monetary policy and financial institutions.

Literature Review and Hypothesis Development

The Big Four banks comprise approximately 70% of the Australian financial market (Canstar 2025), as such, the scope of the examined literature should target the banking sector as a whole, rather than placing its focus on specific portions of the sector.

A multitude of studies have been conducted in regard to the relationship between monetary policy and bank risk. Literature found primarily utilizes the cash rate at a given time to measure the impact of monetary policy on bank risk, with the key difference across studies being the geographical location of the assessed banks. This relationship was often measured through an OLS regression, incorporating factors such as GDP growth and Return on Asset ratios as a means of reducing the error term. This research provides a strong basis for the expected outcome of the findings in this paper, suggesting that geographical variance plays little role in the research, and that bank risk is expected to rise with the tightening of monetary policy.

Studies found that a loose monetary policy, that is; lowering interest rates in an attempt to boost spending and borrowing, was linked to an increase in bank risk-taking. Altunubas, Gambacorta, and Marquz-Ibanez (2010) argue that lower returns on investment caused by a loose monetary policy may increase incentive for a bank to take on more risk for behavioral, contractual, or institutional reasons. More recent studies support this claim, with a study on 119 Chinese banks reaching the same conclusion (Li, Deng, Cui & Pan 2025), with research done on Western Balkan banks further

supporting the existence of a negative correlation between monetary policy rates and bank risk-taking (Gashi & Fetai 2023).

Measurements of the relationship between changes in monetary policy and bank risk levels are not well researched, though relevant sources found further promote that a looser monetary policy increases bank risk-taking. Policy-induced interest rate increases were found to prompt banks into utilizing more expensive sources of funding across banks in the US (English, Van den Heuvel, Zakrajsek 2018). More recently, banks in India were assessed to be more prone to risk upon expansion of monetary policy, however, the extent of impact varied based on bank ownership structure (Ghosh 2025).

In researching the effect of monetary policy on bank lending, Kashyap and Stein (2000) found that banks with a lower ratio of securities to assets faced greater impact in changes of monetary policy. Recent findings further suggest this theme, with a rise in the Fed Funds rate, the US equivalent of the overnight cash rate, increasing earnings more for banks who exhibit a larger difference in the sensitivities of their assets and liabilities (Gomez, Landier, Sraer & Thesmar 2021). However, due to the oligopoly held on the financial institution sector, as well as the similarity in size and capital-to-asset ratio held by the 4 banks examined in this paper, these findings are unlikely to influence the research objective.

Abiding by the findings showed in the reviewed research, we form the following hypothesis: due to the reduced profitability, and higher demand for lending and borrowing, a lower cash rate level is expected to heighten the risk undertaken by banks in an attempt to maintain stability, while the opposite is assumed to be observed at higher cash rate levels. This sentiment is more clearly outlined in the null-hypothesis statement below.

H_0 = *Risk undertaken by the big four Australian banks is not in any way correlated with the level of the cash rate.*

H_1 = *The level of risk undertaken by a bank is influenced by the level of the cash rate, with lower cash rate values leading to increased bank risk, and the opposite being true of higher cash rate values.*

Methodology

To analyse the provided data, a python script was written leveraging a number of python libraries such as matplot-lib, and scikit-learn. The script was used to provide descriptive statistics on the data, as well as to generate visualizations such as a correlation heatmap, an OLS regression with the Cash rate level as the MP_t variable, a line plot showing Z-score trends by bank, line and scatter plots of the Predicted vs Actual Z-score and the Residuals vs Predicted Z-score, and a regression table showing the key

information obtained from the OLS regression. The resulting OLS regression was used to derive an equation fitting that of the suggested empirical model:

$$Z - \widehat{score}_{i,t} = 5.45 + 15.06 \cdot Cash\ rate_t - 0.45 \cdot Size_{i,t} + 3.13 \cdot Capital\ ratio_{i,t} - 0.67 \cdot Asset\ growth_{i,t} - 2.45 \cdot Cost\ to\ income_{i,t} - 0.09 \cdot Non\ interest\ income_{i,t} + 0.01 \cdot GDP\ growth_t + 0.01 \cdot Inflation_t$$

The OLS regression shows the Cash Rate vs Bank Risk by bank to account for unmeasured bank-specific factors such as management practices, and market outlook, in order to isolate the effect of explanatory variables, thus improving the model accuracy. Model fit was further evaluated through an R-squared measurement. Generative AI was used in the process to ensure the script fit coding best-practices, and was an accurate method of analyzing the data. Note that Z-score is an inverse measure of a bank's default risk; the higher the Z-score, the more risk-averse a given bank is.

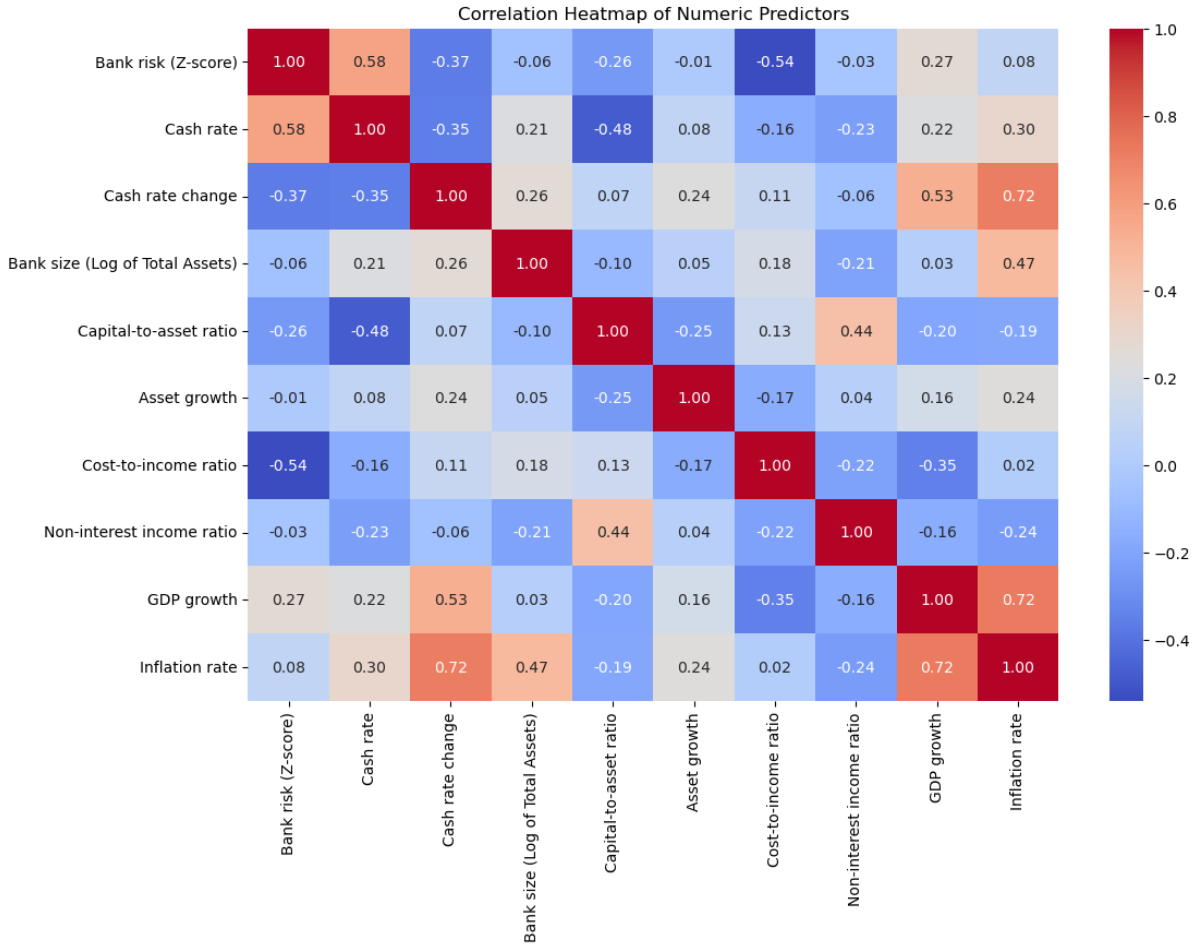
Analysis of Results

For the purpose of examining only the factors used in the OLS equation, the variables "Bank", "Bank ID", and "Year" were removed when creating the table of descriptive statistics. The analysis measured a mean Bank risk of ~2.62, with a somewhat narrow range ([1.33, 3.05]), indicating moderate variability in risk across the 4 banks. The cash rate mean was calculated to be ~0.02, suggesting that monetary policy loosening and tightening was done in a balanced manner across the 12-year period. However, the variable's wide range ([0.00, 0.04]) showcases the dynamic nature of monetary policy. The added influence of the other variables supports the hypothesis that when monetary policy is eased banks are likely to lower risk-aversion due to lower profitability.

	Cash rate	Cash rate change	Bank risk (Z-score)	Bank size (Log of Total Assets)	Capital-to-asset ratio	Asset growth	Cost-to-income ratio	Non-interest income ratio	GDP growth	Inflation rate
count	48.00	47.00	48.00	48.00	48.00	48.00	48.00	48.00	48.00	48.00
mean	0.02	0.00	2.06	5.97	0.06	0.05	0.47	0.39	2.37	2.62
std	0.01	0.01	0.35	0.06	0.01	0.05	0.06	0.21	1.04	1.67
min	0.00	-0.02	1.33	5.85	0.05	-0.19	0.39	-0.58	-0.12	0.85
25%	0.01	-0.00	1.90	5.93	0.06	0.02	0.43	0.31	2.15	1.58
50%	0.02	-0.00	2.09	5.97	0.06	0.05	0.46	0.44	2.44	2.17
75%	0.03	0.00	2.24	6.01	0.07	0.08	0.50	0.51	2.79	2.58
max	0.04	0.03	3.05	6.10	0.08	0.15	0.75	0.64	4.24	6.59

(Figure 1: Descriptive Statistics of the Numerical Variables, Excluding "Bank ID", and "Year")

The correlation matrix illustrates the individual relationships of the numerical factors. Most notably, bank risk exhibits a moderately positive correlation with the cash rate level (0.58), further supporting the hypothesis that monetary policy easing results in an increase in risk-taking behaviour in banks. Conversely, Bank size, the capital-to-asset ratio, cost-to-income ratio and asset growth were all negatively correlated with bank risk (-0.06, and -0.26, -0.52, and -0.01 respectively), implying that the larger a bank's market reach gets the more risk it may take on, with operational inefficiency, and aggressive growth acting as further factors in this idea. GDP growth and inflation each had weak correlation to bank risk (0.27 and 0.08 respectively), indicating that macroeconomic factors play secondary roles compared to bank specific ones when determining risk.

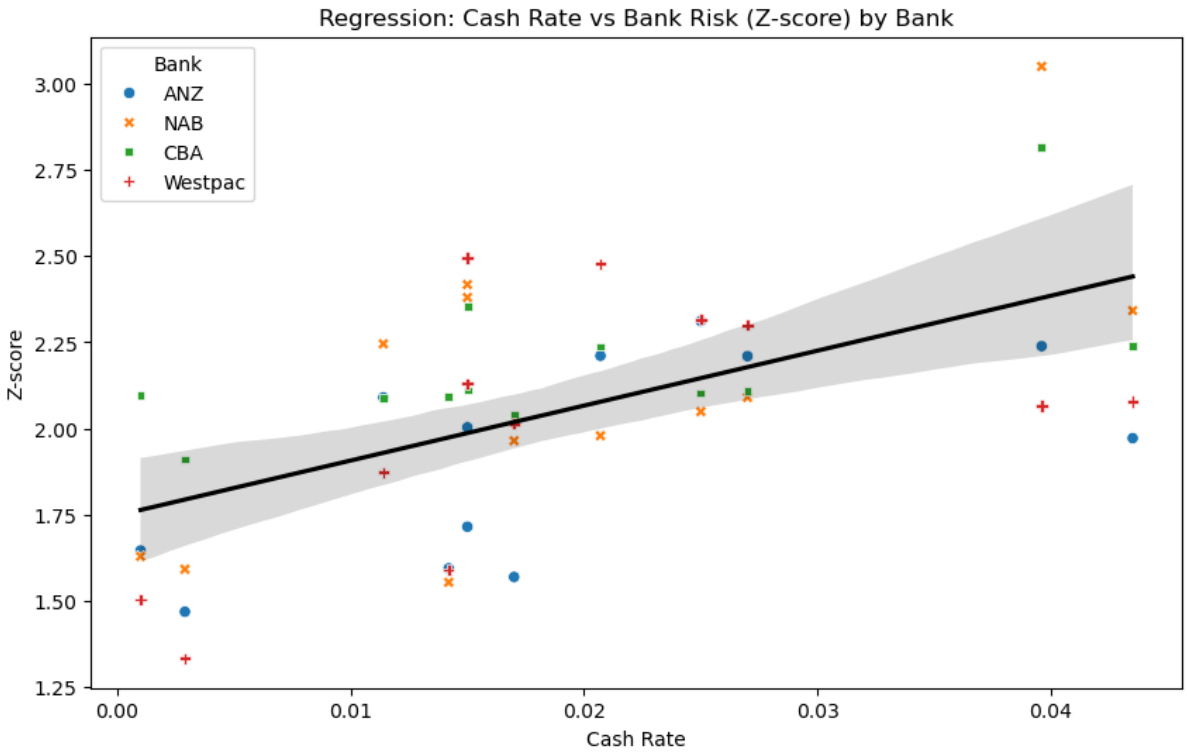


(Figure 2: Correlation Matrix Showing the Individual Relationship Between Numerical Factors)

The OLS regression equation exhibited a positive slope, indicating that as the cash rate level increases (i.e. monetary policy is tightening) bank risk decreases, further evidencing that a loosening of monetary policy results in heightened bank risk. The β coefficient had a value of 15.06, suggesting that a positive increase in the cash rate by one basis point heightens a bank's risk Z-score by 0.1506, ceteris paribus. Other

country-wide factors had relatively low impact on bank risk, with GDP Growth and Inflation rate each having coefficient values of 0.01. Moreover, the bank-specific factors each had coefficient values between -1 and 1, with the exception of the cost-to-income ratio which displayed a value of -2.45, implying that banks with better operating efficiency are more risk-averse in contrast to less efficient ones, this is idea is validated by the variable's low p-Value of 0.0018, indicating statistical significance.

The implied trends can be found across the 4 institutions, although ANZ and Westpac appeared to be more sensitive to rate changes than CBA and NAB. The clustering of Z-score values around the ~0.015 cash rate value indicates that banks are more sensitive to loosening of monetary policy, further supporting the correlation between the two. The variability seen in each bank's Z-score values can be attributed to differences in portfolio structure, as well as company and management structure.

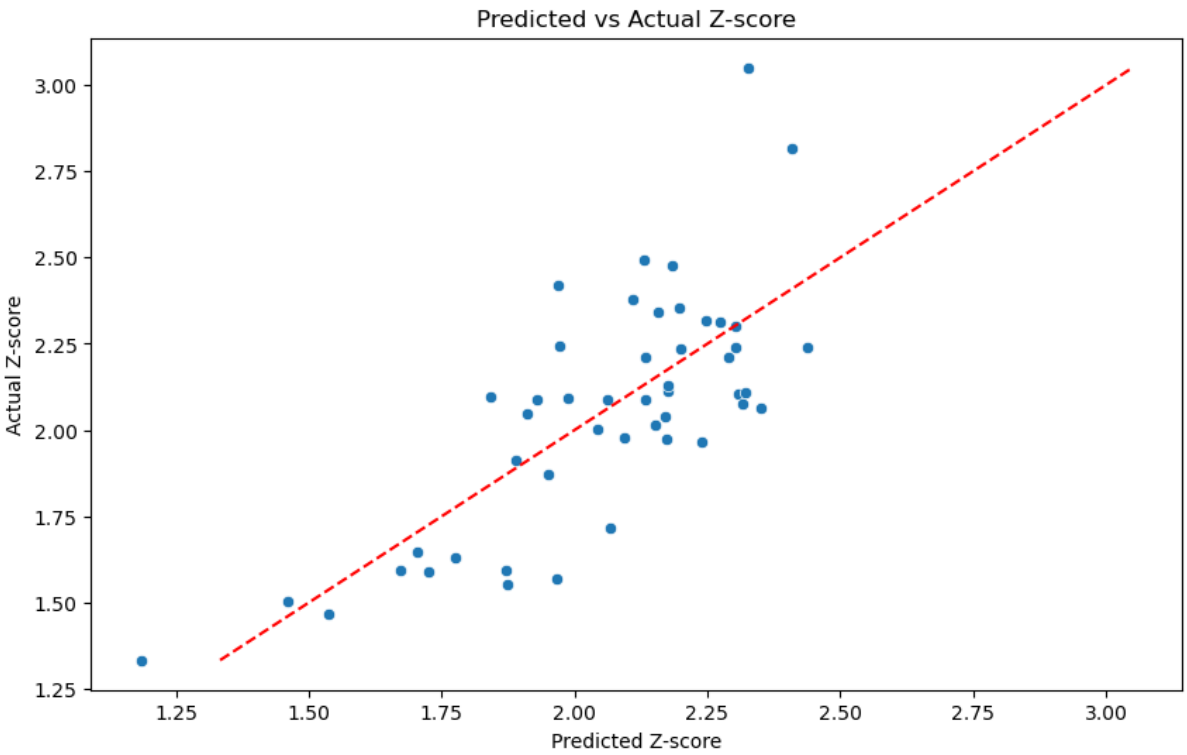


(Figure 3: OLS Regression Showing Cash Rate vs Bank Risk (Z-score) by Bank)

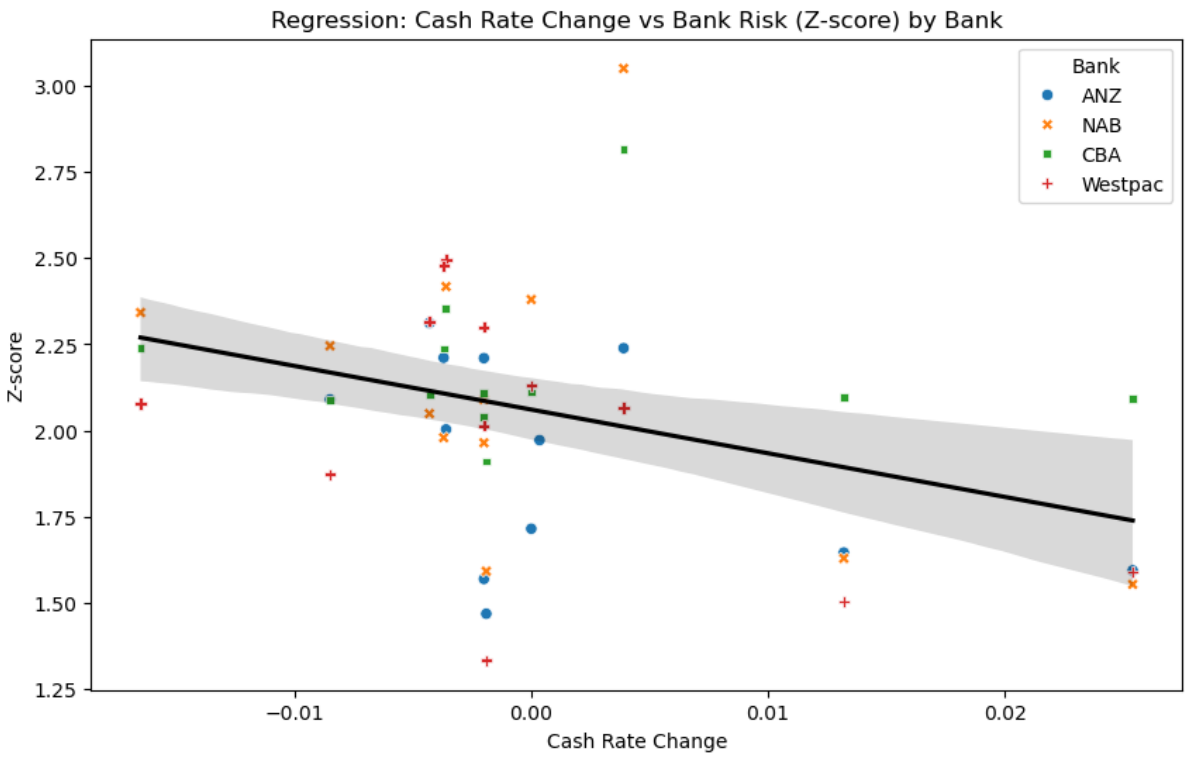
Variable	Coefficient	t-Statistic	p-Value
const	5.4499	1.1422	0.2603
Cash rate	15.0582	4.2445	0.0001
Bank size (Log of Total Assets)	-0.448	-0.5735	0.5696
Capital-to-asset ratio	3.1275	0.3937	0.6959
Asset growth	-0.6727	-0.9104	0.3682
Cost-to-income ratio	-2.4461	-3.3571	0.0018
Non-interest income ratio	-0.0851	-0.3921	0.6971
GDP growth	0.011	0.1554	0.8773
Inflation rate	-0.0062	-0.1303	0.897
R-squared	0.5615	-	-

(Figure 4: Regression Table Summary of Key Values)

Acting as a means of validating the model, the "Predicted vs Actual Z-score" graph shows the OLS regression performing with high accuracy when investigating the aim of this study. Most of the data points lie near the line of $y=x$, suggesting that predicted Z-score values closely match the actual data. Additionally, the model had an R2 value of 0.5615, suggesting that 56.15% of the risk a bank exhibits can be explained by the model. Interestingly, an OLS regression that used the *Cash Rate Change* as the MP_t variable was found to have an R2 value of 0.6095, suggesting that the degree of cash rate change may be more impactful on bank risk than the simple level of cash rate.

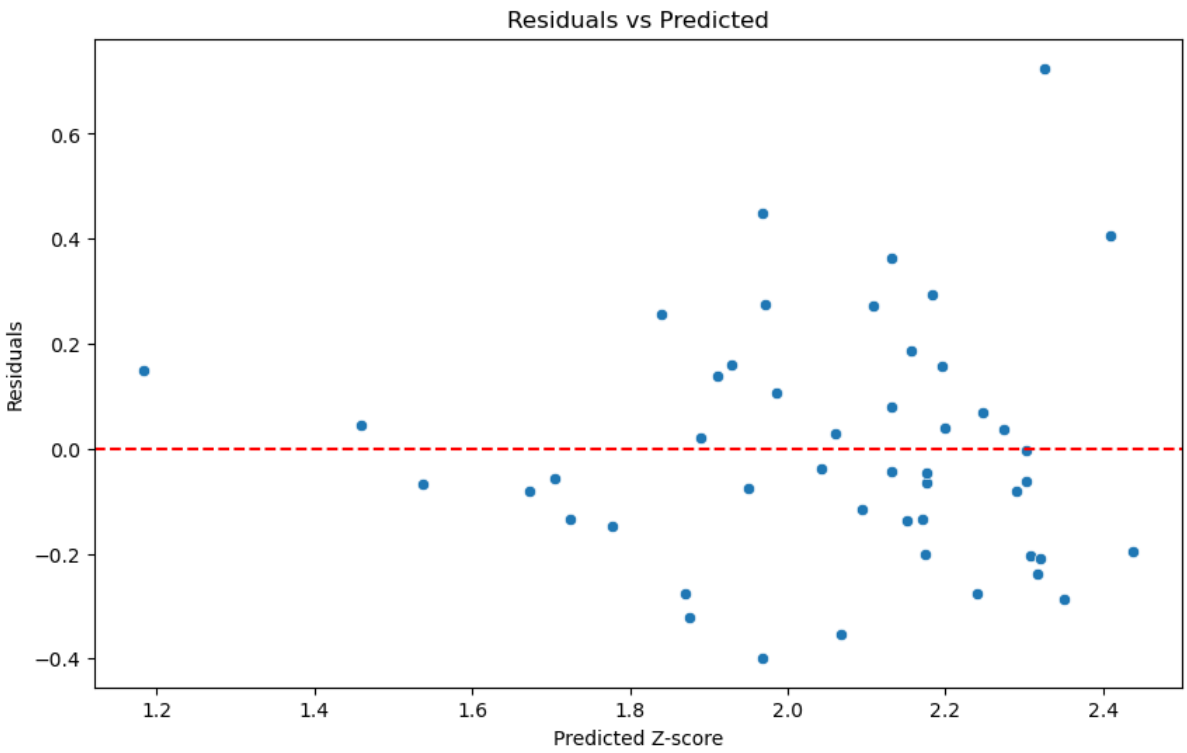


(Figure 5: Predicted vs Actual Z-score)



(Figure 6: OLS Regression Showing Cash Rate Change vs Bank Risk (Z-score) by Bank)

To further validate the model's reliability a scatterplot showing the residual values vs the predicted values was constructed. The residuals in the graph were randomly scattered, following no identifiable pattern, and displaying no heteroskedasticity. The displayed randomness suggests that the model does not over- or under-predict bank risk at any level. The residual values showed some tight clustering around ±0.3 area, indicating that prediction errors were evenly distributed and insignificant. This chart supports the assumption of homoskedasticity, and further indicates that the model is of the appropriate nature.



(Figure 7: Residuals vs Predicted Values)

The combination of the above graphs and observations provides a consistent narrative: tightening of monetary policy through negative changes in the cash rate is associated with an increase in risk undertaken by the big 4 Australian banks. This relationship is most clearly encapsulated by the OLS regression model. The findings within this study serve as a reminder to policy makers to consider the economy-wide implications of policy changes, in order to avoid unwanted financial consequences.

Recommendations

Based on the above findings, it is recommended that both policymakers and those involved in the financial sector closely consider the behavioural response in risk taking by banks, as a result of changes in the cash rate. Financial institutions may enhance internal risk management tools, and strive for better portfolio management that reflects the level of risk willing to be undertaken. Future research on this topic should explore the impact of less conventional monetary tools on bank risk, as well as the effects exhibited by individual bank assets in times of monetary policy change.

Conclusion

This study has examined the impact had by changes in monetary policy on the level of risk undertaken by the four largest banks in the Australian financial sector. Findings suggest that loosening of monetary policy through negative changes in the cash rate by the RBA are associated with an increase in the risk undertaken by banks, whereas tightened monetary policy has the opposite effect, supporting the formulated

hypothesis. These insights showcase the importance of meeting macroeconomic targets, whilst simultaneously considering the wider implications of policy changes.

References

Altunbas, Y., Gambacorta, L., & Marques-Ibanez, D. (2010). Does Monetary Policy Affect Bank Risk-Taking? SSRN Electronic Journal, 298. https://doi.org/10.2139/ssrn.1572798

Bajaj, V., & Pridmore, J. (2025, June 30). Compare The Big Four Banks in Australia. Retrieved September 1, 2025, from Canstar website: https://www.canstar.com.au/home-loans/compare-the-big-four-banks-in-australia/

Gashi, F., & Fetai, B. (2023). The Impact of Monetary Policy on Bank Risk in the Western Balkan Countries. SEEU Review, 18(2), 4–18. https://doi.org/10.2478/seeur-2023-0016

Gomez, M., Landier, A., Sraer, D., & Thesmar, D. (2020). Banks' exposure to interest rate risk and the transmission of monetary policy. Journal of Monetary Economics, 117, 543–570. https://doi.org/10.1016/j.jmoneco.2020.03.011

Li, Y., Deng, M., Cui, Y., & Pan, J. (2025). Does monetary policy impact bank risk-taking? An empirical study based on the data of 119 commercial banks in China. Finance Research Letters, 81, 107547. https://doi.org/10.1016/j.frl.2025.107547

Stein, J., & Kashyap, A. (2000). What Do A Million Observations on Banks Say About the Transmission of Monetary Policy? | Jeremy Stein. American Economic Review, 90, 407–428. Retrieved from https://stein.scholars.harvard.edu/publications/what-do-million-observations-banks-say-about-transmission-monetary-policy